Part X — Applied Physics · Chapter 55

Physics of Earth and Atmosphere

How mechanics, fluids, heat transfer, waves, and radiation explain Earth's systems

> *Hurricanes are heat engines driven by the temperature difference between the warm ocean surface and the cold upper atmosphere. Earthquakes propagate as waves through rock, revealing the structure of the deep interior by how they refract and reflect off density boundaries. The greenhouse effect is a consequence of molecular absorption spectra — CO_2 and water vapour absorb infrared radiation more strongly than visible light. The tides arise from the gradient of the Moon's gravitational field across Earth's diameter. None of these phenomena requires new physics — they follow from the mechanics, thermodynamics, wave physics, fluid dynamics, and electromagnetism already covered in this course. This chapter applies those principles to Earth as a physical system: a rocky sphere surrounded by a thin layer of gas, bathed in radiation from the Sun, heated from within by radioactive decay, and shaped by four billion years of fluid and tectonic motion.*

55.1 Earth as a Physical System

Earth is a layered physical system driven by two energy sources: solar radiation received at the surface and radioactive decay heat released from the interior. The interplay of these energy inputs with Earth's mass, rotation, and surface properties produces the atmosphere, oceans, weather, climate, and geological activity we observe.

Earth's interior is divided into four main layers by composition and physical state. The inner core (radius $\approx$1220 km) is solid iron-nickel at extreme pressure ($\approx$3.6 Mbar) and temperature ($\approx$5000–6000 K). The outer core (1220–3480 km radius) is liquid iron-nickel and generates Earth's magnetic field through convective motion. The mantle (3480–6371 km from centre, depth 30–2890 km from surface) is solid rock that behaves plastically over geological timescales — it convects on timescales of millions of years, driving plate tectonics. The crust (0–70 km depth) is thin solid rock: oceanic crust ($\approx$7 km thick, density $\approx$3000 kg/m^3) and continental crust (30–70 km thick, density $\approx$2700 kg/m^3).

The atmosphere is a thin shell of gas held by gravity. Total mass $\approx 5.15 \times 10^{18}$ kg, but only about 50% lies below 5.5 km altitude and 99% below 32 km. The atmosphere is divided into layers by temperature profile: troposphere (0–12 km, temperature decreasing with altitude), stratosphere (12–50 km, temperature increasing due to ozone absorption of UV), mesosphere (50–80 km, temperature again decreasing), and thermosphere/ionosphere (above 80 km, rapidly increasing temperature due to direct absorption of extreme UV and X-rays).

Layer	Altitude Range	Temperature Trend	Key Process
Troposphere	0–12 km	Decreases with altitude	Weather, cloud formation, convection
Stratosphere	12–50 km	Increases with altitude	Ozone absorbs UV radiation
Mesosphere	50–80 km	Decreases with altitude	Meteors burn up here
Thermosphere	80–700 km	Increases with altitude	Aurora, satellite orbits
Exosphere	>700 km	Decreasing particle density	Gradual transition to space

Table 55.1 *Atmospheric layers and their key physical characteristics.*

55.2 Gravity, Mass, and Earth's Shape

Earth is not a perfect sphere. Rotation causes an equatorial bulge: the equatorial radius (6378 km) is 21 km greater than the polar radius (6357 km). This shape is called an oblate spheroid. The bulge arises because rotating material must have sufficient centripetal acceleration, which reduces the effective gravitational acceleration at the equator.

Surface gravitational acceleration g varies with latitude and altitude. At the equator, $g \approx 9.780$ m/s²; at the poles, $g \approx 9.832$ m/s². The difference (about 0.5%) has two causes: the poles are closer to Earth's centre (stronger gravity) and there is no centripetal acceleration reduction at the poles (since the centripetal acceleration needed for circular motion is zero at the pole). Altitude reduces g according to $g(h) = g_0(R/(R+h))^2$, where R is Earth's radius.

$$g(h) = g_0 \left(R_E / (R_E + h)\right)^2$$

Variation of gravitational acceleration with altitude h above Earth's surface. $g_0 = 9.81$ m/s² at sea level; $R_E = 6.371 \times 10^6$ m. At 400 km (ISS altitude), $g \approx 8.69$ m/s² — not zero. Astronauts *experience weightlessness because they are in free fall, not because $g = 0$.*

The gravitational potential energy of the atmosphere, ocean, and surface materials drives geophysical processes. Water falls from elevation to sea level, releasing potential energy that drives rivers, erosion, and hydroelectric power. Air descends from high pressure to low pressure, converting potential energy to kinetic energy (wind). Rock collapses in landslides and calves from cliffs when gravitational force exceeds material strength.

Example — g at ISS altitude The ISS orbits at h = 408 km. $g(408\text{ km}) = 9.81 \times (6371 / (6371 + 408))^2 = 9.81 \times (6371/6779)^2 = 9.81 \times 0.8838 = 8.67\text{ m/s}^2$. Gravity at the ISS is about 88% of surface gravity. The crew is weightless because the station is in free fall (orbit), not because gravity is absent.

55.3 Atmospheric Pressure and Density

Atmospheric pressure at any altitude equals the weight per unit area of the column of air above that altitude. As altitude increases, less air is overhead, so pressure decreases. The exact relationship depends on how temperature varies with altitude, but for an atmosphere at approximately constant temperature (isothermal), pressure decreases exponentially:

$$P(h) = P_0 e^{(-h/H)}$$

Barometric formula for an isothermal atmosphere. $P(h)$ = pressure at altitude h; P_0 = sea-level pressure (101 325 Pa); H = scale height = $RT/(Mg) \approx 8.5$ km for Earth's troposphere ($R = 8.314$ J/mol·K, $T \approx 250$ K mean, $M = 0.029$ kg/mol, $g = 9.81\text{ m/s}^2$). Pressure halves approximately every 5.5 km.

Because density $\rho = PM/(RT)$ (ideal gas law), density decreases with altitude at the same exponential rate as pressure in an isothermal atmosphere. At 10 km altitude, air density is about 34% of sea-level density — which is why aircraft need pressurised cabins and mountaineers carry supplemental oxygen at altitudes above ≈8000 m (the "death zone" where the partial pressure of oxygen is insufficient to sustain human life without supplementation).

Density also varies with temperature and humidity at fixed altitude. Warm air is less dense than cold air (ideal gas: lower ρ at higher T for fixed P). Humid air is less dense than dry air (water vapour, M = 18 g/mol, is lighter than N_2, M = 28 g/mol, and O_2, M = 32 g/mol). These density differences drive convection and weather.

Example — Pressure at 5500 m $P(5500) = 101325 \times e^{(-5500/8500)} = 101325 \times e^{(-0.647)} = 101325 \times 0.524 = 53\,094\text{ Pa} \approx 53.1\text{ kPa}$. At 5500 m, atmospheric pressure is roughly half of sea-level pressure, consistent with the empirical rule that pressure halves approximately every 5.5 km.

55.4 Heat Transfer in the Atmosphere

The atmosphere exchanges heat with the surface and with space by all three modes of heat transfer: radiation (dominant), convection (responsible for weather), and conduction (significant only at the immediate surface-atmosphere interface).

The surface absorbs solar radiation (mostly visible, 0.4–0.7 μm) and warms. It then re-emits energy as infrared radiation (peak wavelength ≈9–10 μm for a surface at 300 K, from Wien's law λ_max = b/T). Atmospheric greenhouse gases (H_2O, CO_2, CH_4, N_2O) absorb this outgoing infrared and re-radiate it in all directions, including back toward the surface — the greenhouse effect. Without any greenhouse gases, Earth's mean surface temperature would be about −18°C; the observed mean is about +15°C, a difference of 33°C attributable to the natural greenhouse effect.

$$\lambda_{max} = b / T \qquad \text{(Wien's displacement law)}$$

Wien's displacement law. λ_max = wavelength of peak emission (m); b = Wien's constant = 2.898 × 10^{-3} m·K; T = absolute temperature (K). The Sun (T ≈ 5778 K) emits peak radiation at λ ≈ 0.5 μm (visible green); Earth's surface (T ≈ 288 K) emits peak radiation at λ ≈ 10 μm (infrared).

The lapse rate is the rate at which temperature decreases with altitude in the troposphere. The average environmental lapse rate is about 6.5°C/km. The dry adiabatic lapse rate (DALR) is 9.8°C/km — the rate at which a parcel of unsaturated air cools as it rises and expands adiabatically. When the environmental lapse rate exceeds the DALR, the atmosphere is unstable to convection: a rising air parcel finds itself warmer (less dense) than its surroundings and continues to rise. This is the condition for thunderstorm development.

55.5 Convection and Weather Systems

Convection is the dominant mechanism of vertical heat transport in the troposphere. Warm, less-dense air rises (buoyancy); cooler, denser air sinks. Rising air expands and cools adiabatically; if it cools to the dew point, water vapour condenses, releasing latent heat (2.5 MJ/kg), which partially offsets the adiabatic cooling and allows the parcel to continue rising. This is how cumulonimbus (thunderstorm) clouds develop.

Large-scale convection organises into circulation cells. The Hadley cell: air heated at the equator rises, moves poleward at altitude, descends at about 30° latitude (creating subtropical deserts), and returns equatorward as surface trade winds. The Ferrel cell

(30–60°) and Polar cell (60–90°) complete the three-cell model of each hemisphere's general circulation.

Low-pressure systems (cyclones) develop where surface air rises (warm, moist, or converging air). Air spirals inward and upward due to the Coriolis effect — counterclockwise in the Northern Hemisphere, clockwise in the Southern Hemisphere. As air rises, it cools and condenses, producing clouds and precipitation. High-pressure systems (anticyclones) have descending air, diverging outward at the surface; descending air warms and dries, producing clear skies.

Fronts are boundaries between air masses of different temperature and density. At a cold front, advancing dense cold air undercuts warm air, forcing it rapidly upward — producing vigorous convection, thunderstorms, and heavy but brief precipitation. At a warm front, warm air gradually overrides retreating cold air along a gentle slope, producing extensive stratiform cloud and prolonged light precipitation.

> **Historical Note:** The Coriolis effect, identified by Gaspard-Gustave de Coriolis in 1835, arises because Earth rotates beneath moving air masses. A northward-moving air parcel in the Northern Hemisphere deflects eastward because Earth's rotational surface speed decreases with latitude — the air retains the higher eastward velocity of its origin point. The effect scales with latitude (zero at equator, maximum at poles) and wind speed, and is responsible for the spiral structure of hurricanes, mid-latitude storm systems, and ocean gyres.

55.6 Wind, Pressure Gradients, and Fluid Motion

Wind is horizontal air motion driven by pressure gradient forces. Air flows from high to low pressure (down the pressure gradient), but is deflected by the Coriolis effect and friction. In the free atmosphere (above the boundary layer), the balance between the pressure gradient force and the Coriolis force produces geostrophic wind: wind that blows parallel to isobars (lines of equal pressure) at a speed proportional to the pressure gradient.

$$v_g = (1/f\rho) \times (dP/dn)$$

Geostrophic wind equation. v_g = geostrophic wind speed (m/s); f = Coriolis parameter = $2\Omega \sin\varphi$ ($\Omega = 7.27 \times 10^{-5}$ rad/s, φ = latitude); ρ = air density (kg/m³); dP/dn = pressure gradient perpendicular to the wind (Pa/m). Tighter isobar spacing means a stronger pressure gradient and faster wind.

The jet stream is a narrow band of fast geostrophic wind at the tropopause (8–12 km altitude), where the temperature contrast between polar and tropical air masses is greatest. Wind speeds of 100–300 km/h are common; the jet stream steers mid-latitude weather systems and has a major effect on transcontinental aviation flight times.

Bernoulli's equation applies to air flow as well as liquid flow. Air flowing over a hill or building accelerates (continuity: narrowing flow cross-section), reducing pressure (Bernoulli). This produces low pressure zones on the downwind side of buildings, reduced pressure on the upper surface of aircraft wings (lift), and the Venturi effect in carburettors and spray paint guns. Wind turbines extract kinetic energy from the flow; the Betz limit (16/27 ≈ 59.3% of wind kinetic energy) is the theoretical maximum fraction extractable by any wind turbine.

55.7 Water Cycle Physics

The water (hydrological) cycle describes the continuous movement of water through Earth's systems: evaporation from oceans and land surfaces, atmospheric transport, condensation and precipitation, and return flow via rivers and groundwater. The cycle is driven by solar energy and gravity.

Evaporation requires latent heat (L_v = 2.5 MJ/kg at 20°C). The ocean absorbs solar energy and evaporates water; this transfers latent heat from the ocean surface to the atmosphere where it is released upon condensation. The global mean evaporation rate is about 1 m/year from the ocean surface, transferring $\approx 2.5 \times 10^{23}$ J/year to the atmosphere — about six times more energy than the atmosphere receives directly by absorbing solar radiation. The water cycle is therefore the dominant mechanism of energy transport between the surface and the atmosphere.

Precipitation requires condensation nuclei (tiny aerosol particles on which water vapour condenses) and supersaturation. Raindrops fall at their terminal velocity, where drag equals weight: $F_{drag} = \frac{1}{2}\rho_{air} C_D A v^2 = mg$. A typical raindrop of radius 1.5 mm reaches a terminal velocity of about 6–7 m/s. Larger drops fall faster but eventually break apart due to aerodynamic instability; maximum stable raindrop radius is about 4 mm.

$$v_t = \sqrt{2mg / (\rho_{air} C_D A)}$$

Terminal velocity of a falling raindrop. m = mass (kg); g = 9.81 m/s²; ρ_{air} = 1.2 kg/m³; $C_D \approx 0.47$ (sphere); A = cross-sectional area (m²). In practice, raindrops deform from spherical, modifying C_D at large sizes.

Snow forms when ice crystals grow directly from water vapour (deposition) around condensation nuclei at temperatures below 0°C and below the frost point. Ice crystals grow into hexagonal plates, dendrites, columns, or needles depending on temperature and humidity. Snowflakes are aggregates of many crystals; hail forms when raindrops are lofted repeatedly through the freezing level by updrafts in severe thunderstorms, accumulating concentric layers of ice.

55.8 Ocean Currents and Waves

Ocean currents are driven by two mechanisms: surface currents driven by wind stress on the ocean surface, and thermohaline (deep) circulation driven by density differences arising from temperature and salinity variations. Surface currents form large gyres (rotating circulation systems) in each ocean basin, driven by prevailing winds and deflected by the Coriolis effect and continental boundaries. The Gulf Stream carries warm tropical water northward along the eastern coast of North America at flow rates of ∶ 30–50 Sv (sverdrups; 1 Sv = 10^6 m^3/s).

Thermohaline circulation (the global ocean conveyor belt) is driven by the sinking of cold, salty (dense) water in the North Atlantic and around Antarctica. Cold deep water flows southward and eventually upwells in the Indo-Pacific, completing a circuit with a turnover time of roughly 1000 years. This circulation transports massive amounts of heat poleward and plays a critical role in regulating regional climates, particularly in Western Europe.

Ocean surface waves are generated by wind transferring energy to the water surface. The physics of waves (Chapter 22) applies directly: wave speed $v = \sqrt{g\lambda/(2\pi)}$ for deep-water gravity waves, where λ is wavelength. Longer waves travel faster: a swell of wavelength 200 m travels at $v = \sqrt{9.81 \times 200/(2\pi)} \approx 17.7$ m/s. As waves enter shallow water (depth < $\lambda/2$), they slow, their wavelength decreases, amplitude increases, and eventually they break when amplitude/depth ≈ 0.8 (the breaking criterion).

> **Example — Deep water wave speed** A storm swell has wavelength 150 m. Wave speed $v = \sqrt{g\lambda / 2\pi} = \sqrt{9.81 \times 150 / 6.283} = \sqrt{234.1} \approx 15.3$ m/s. The wave period $T = \lambda/v = 150/15.3 \approx 9.8$ s. These are consistent with typical ocean swell observed hundreds of kilometres from storms.

Tsunamis are not wind-driven surface waves but long-wavelength gravity waves generated by submarine earthquakes, volcanic eruptions, or landslides. In the open ocean, water depth $d \ll \lambda$, so they travel as shallow-water waves at $v = \sqrt{gd}$. In the Pacific, mean depth $\approx$4000 m, giving tsunami speeds of $\sqrt{9.81 \times 4000} \approx 198$ m/s $\approx$ 714 km/h. Amplitude in deep water is small ($\approx$0.5 m); as tsunamis enter shallow coastal water, they slow dramatically and energy conservation forces amplitude to increase, potentially to 10–30 m at the coast.

55.9 Seismic Waves and Earth’s Interior

Earthquakes generate seismic waves that propagate through Earth and can be detected worldwide by seismographs. There are two main types. Body waves travel through the interior: P-waves (primary, compressional — particle motion parallel to propagation direction, travel through solids and liquids) and S-waves (secondary, shear — particle motion perpendicular to propagation direction, travel through solids only). Surface waves (Rayleigh and Love waves) travel along Earth's surface and cause most of the damage in earthquakes.

$$v_P = \sqrt{(K + 4G/3) / \rho} \qquad v_S = \sqrt{G / \rho}$$

Seismic wave speeds. K = bulk modulus (resistance to compression); G = shear modulus (resistance to shear); ρ = density. P-waves travel faster than S-waves in all materials. S-waves cannot propagate through liquids (G = 0 for fluids), which is how the liquid outer core was first identified.

The time difference between the arrival of P-waves and S-waves at a seismograph station gives the distance to the earthquake epicentre. With three stations, the epicentre can be located by triangulation. The magnitude of an earthquake is measured on the moment magnitude scale (M_w), which is logarithmic: each integer step represents approximately 32 times more energy release. An M 8.0 earthquake releases about 1000 times more energy than an M 6.0 earthquake.

By analysing the travel times and paths of seismic waves from thousands of earthquakes, geophysicists have constructed a detailed three-dimensional model of Earth's interior. The existence of the liquid outer core was identified by the S-wave shadow zone (a region where no direct S-waves arrive, because S-waves cannot propagate through liquid). The solid inner core was inferred from reflected P-waves. The crust-mantle boundary (Mohorovičić discontinuity, or Moho) was identified from the refraction of seismic waves at the density change between crustal rock and denser mantle rock.

> **Key Point:** The Mohorovičić discontinuity (Moho) was discovered in 1909 by Croatian seismologist Andrija Mohorovičić, who noticed that seismic waves from earthquakes arrived at some stations faster than expected, and correctly inferred they had refracted at a deep boundary into a higher-velocity layer — the top of the mantle. The Moho lies at about 30–70 km depth under continents and 7 km under ocean basins.

55.10 Earth's Magnetic Field

Earth behaves as a magnetic dipole with field strength ≈25–65 μT at the surface. The field is generated by convective motion of liquid iron in the outer core — the geodynamo. Fluid motion driven by heat and compositional buoyancy carries electrical currents that generate the magnetic field through electromagnetic induction, in a self-sustaining feedback (a natural dynamo). The geodynamo is complex and chaotic: the field reverses polarity irregularly, with the last reversal (Brunhes-Matuyama) about 780 000 years ago. Several hundred reversals are recorded in the palaeomagnetic record of oceanic rocks.

Earth's magnetic field is tilted about 11° from the rotation axis, so magnetic north and geographic north do not coincide. The difference — magnetic declination — varies with location and changes over time as the field evolves. Navigators have accounted for declination for centuries; modern GPS has reduced its practical importance for most applications, but it remains critical for magnetic compass navigation and geological mapping.

The magnetosphere is the region around Earth dominated by its magnetic field rather than the solar wind (a stream of charged particles from the Sun). The magnetosphere deflects most of the solar wind, protecting the surface from direct particle bombardment. Charged particles (mainly protons and electrons) trapped in the Van Allen radiation belts follow helical paths along field lines. Where field lines converge near the poles, particles spiral inward, collide with atmospheric atoms, and produce the aurora (borealis in the north, australis in the south).

Rock formed at mid-ocean ridges records the polarity of Earth's magnetic field at the time of solidification (palaeomagnetism). Symmetric stripes of normal and reversed magnetisation on either side of mid-ocean ridges provided crucial evidence for seafloor spreading and, ultimately, the theory of plate tectonics. The spacing of stripes encodes the spreading rate; comparison with the known timescale of reversals provides an independent measure of how fast plates move (typically 1–10 cm/year).

55.11 Solar Radiation and Earth's Energy Balance

The Sun radiates as a near-perfect blackbody at ≈5778 K. Total solar power output (luminosity) $L_\odot = 3.85 \times 10^{26}$ W. The solar irradiance (solar constant) at Earth's mean orbital distance is $S_0 \approx 1361$ W/m² — the power received per unit area of a surface perpendicular to the solar beam at the top of the atmosphere.

Earth intercepts solar radiation over a cross-sectional area of πR_E^2. It re-radiates energy over the full spherical area $4\pi R_E^2$. The planetary albedo $\alpha \approx 0.30$ (30% of incoming solar radiation is reflected). At radiative equilibrium (energy in = energy out):

$$S_0(1-\alpha)\pi R_E^2 = 4\pi R_E^2\,\sigma T_e^4 \quad\Rightarrow\quad T_e = [S_0(1-\alpha)/(4\sigma)]^{1/4}$$

Radiative equilibrium temperature (effective temperature). T_e = effective radiating temperature of Earth; α = albedo (≈0.30); S_0 = solar constant (1361 W/m²); σ = Stefan-Boltzmann constant (5.67×10^{-8} W/m²K⁴). Substituting gives $T_e \approx 255$ K = −18°C. The actual mean surface temperature ≈15°C reflects the additional warming of the greenhouse effect.

The greenhouse effect increases the effective emitting altitude of Earth. Greenhouse gases absorb outgoing infrared and re-radiate at their (lower) temperature; to maintain energy balance, the surface temperature must rise until sufficient radiation escapes from above the gas absorption region. An enhanced greenhouse effect (increased concentration of CO_2, CH_4, N_2O) raises the emitting altitude further, requiring higher surface temperatures to maintain balance — the physical mechanism of anthropogenic climate change.

Solar variability also affects climate. The Sun's output varies by about 0.1% over the 11-year sunspot cycle. Larger variations occur on multi-century timescales; the Maunder Minimum (1645–1715), a period of very low sunspot activity, coincided with the coldest part of the Little Ice Age in Europe. However, solar variability alone cannot explain the warming trend since 1950; changes in greenhouse gas concentrations have been the dominant driver.

55.12 Climate as a Physical System

Climate is the long-term statistical behaviour of the atmosphere-ocean-land system: averages, variability, and extremes of temperature, precipitation, wind, and other meteorological variables over decades. It is distinct from weather, which is the day-to-day state of the atmosphere. The physical climate system is governed by energy balance, fluid dynamics, and feedback mechanisms.

Positive feedbacks amplify initial perturbations. Ice-albedo feedback: warming melts ice (high albedo) and exposes ocean or land (low albedo), reducing reflection and increasing absorbed solar radiation, which causes further warming. Water vapour feedback: warming increases atmospheric water vapour (a powerful greenhouse gas), which causes further warming. These feedbacks roughly double the warming expected from CO_2 alone.

Negative feedbacks dampen perturbations. Planck feedback: as temperature increases, outgoing longwave radiation increases as T^4 (Stefan-Boltzmann), tending to restore equilibrium. Lapse rate feedback (partially negative in the tropics) modifies the vertical temperature structure. Cloud feedbacks are complex and remain the largest source of uncertainty in climate projections.

Climate sensitivity is the equilibrium warming for a doubling of CO_2 concentration. The likely range is 2.5–4.0°C according to the IPCC Sixth Assessment Report (2021–2022), with a best estimate of $\approx$3.0°C. This value is constrained by physical theory, observations of past climate change, and climate model ensembles. The physics is well understood; the uncertainties lie mainly in cloud feedbacks, ocean heat uptake timescales, and ice sheet dynamics.

Study Tip: Climate physics synthesises almost every topic in this course: radiation (solar input, planetary emission), thermodynamics (heat transfer, lapse rate, latent heat), fluid dynamics (atmospheric and ocean circulation), wave physics (seismic waves, ocean waves), and electromagnetism (the geomagnetic field). A strong grasp of these fundamentals makes climate science comprehensible rather than mysterious.

55.13 Common Mistakes in Earth-System Physics

- Thinking Earth's climate is a simple thermostat. The climate system has multiple timescales — atmosphere (days to weeks), ocean mixed layer (months to years), deep ocean (centuries to millennia), and ice sheets (millennia). A forcing change does not produce instantaneous equilibrium; the system has thermal inertia and is still adjusting to past forcing changes.
- Confusing weather and climate. Weather is the atmospheric state on timescales of hours to weeks. Climate is the statistical distribution of weather states over decades. Individual extreme weather events cannot be directly attributed to climate change; changes in the frequency and intensity of events can be.
- Applying the barometric formula without checking the temperature assumption. The formula $P = P_0 e^{(-h/H)}$ assumes constant temperature. Real atmospheric temperature varies with altitude, so the formula is approximate. For large altitude ranges (especially through the stratosphere), a more detailed model is required.
- Forgetting that S-waves cannot propagate through liquids. This is the fundamental evidence for the liquid outer core. Students sometimes assume seismic wave speeds depend only on density and forget that S-waves require a non-zero shear modulus, which is zero for any fluid.
- Mixing up deep-water and shallow-water wave formulas. Deep-water gravity waves: $v = \sqrt{g\lambda/2\pi}$ (speed depends on wavelength). Shallow-water waves (tsunamis, tides): $v = \sqrt{gd}$ (speed depends on depth). The transition occurs when depth $\approx \lambda/2$.
- Assuming the greenhouse effect and the ozone hole are the same problem. The greenhouse effect concerns infrared radiation absorption by CO_2, H_2O, and CH_4 in the troposphere. The ozone hole concerns UV absorption and destruction of stratospheric ozone by halogen radicals (CFCs). These are distinct physical-chemical phenomena with different causes and effects.

Common Mistake: A very common error: the solar constant $S_0 \approx 1361$ W/m² is the power per unit area at the top of the atmosphere on a surface perpendicular to the solar beam. To find Earth's average energy input per unit surface area, divide by 4 (spherical distribution) and multiply by $(1-\alpha)$. The factor of 4 is frequently omitted, leading to an overestimate of solar input by a factor of four.

Chapter Summary

Earth is a layered physical system driven by solar radiation and internal radioactive heating. Gravity, fluid dynamics, thermodynamics, wave physics, and electromagnetism all operate in Earth's interior, surface, oceans, and atmosphere.

- Earth's interior: layered by density and phase; mantle convects over geological time, driving plate tectonics; liquid outer core generates the magnetic field.
- Atmospheric structure: troposphere (weather, lapse rate), stratosphere (ozone, UV absorption), higher layers. Pressure decreases exponentially with altitude (barometric formula).
- Atmospheric heat transfer: solar radiation in (visible), infrared out; greenhouse gases absorb outgoing infrared and warm the surface by ≈33°C above the no-greenhouse baseline of −18°C.
- Convection and weather: instability occurs when environmental lapse rate exceeds DALR; Hadley, Ferrel, and Polar cells drive global circulation; Coriolis effect produces cyclone rotation.
- Wind and pressure gradients: geostrophic wind balances pressure gradient and Coriolis force; jet stream at tropopause steers weather systems.
- Water cycle: latent heat from evaporation is the dominant energy transport from surface to atmosphere; terminal velocity of rain drops; tsunami speeds scale with $\sqrt{gd}$.
- Ocean currents: surface gyres (wind-driven); thermohaline circulation (density-driven); deep water wave speed $v = \sqrt{g\lambda/2\pi}$; tsunami speed $v = \sqrt{gd}$.
- Seismic waves: P-waves (compressional) travel through all media; S-waves (shear) cannot travel through liquids — this reveals the liquid outer core; triangulation locates epicentres.
- Earth's magnetic field: generated by geodynamo in liquid outer core; protects surface from solar wind; records polarity reversals in ocean floor rocks.
- Energy balance: $T_e = [S_0(1-\alpha)/(4\sigma)]^{1/4} \approx 255$ K without greenhouse effect; observed +15°C reflects 33°C of greenhouse warming; enhanced greenhouse effect drives anthropogenic climate change.

Key Terms

lapse rate: The rate of decrease of air temperature with increasing altitude; environmental lapse rate $\approx$6.5°C/km; dry adiabatic lapse rate (DALR) = 9.8°C/km.

barometric formula: $P(h) = P_0 e^{(-h/H)}$; the exponential decrease of atmospheric pressure with altitude in an approximately isothermal atmosphere; scale height H $\approx$8.5 km for Earth.

greenhouse effect: The warming of Earth's surface by atmospheric gases (H_2O, CO_2, CH_4, N_2O) that absorb outgoing infrared radiation and re-radiate it toward the surface.

Coriolis effect: The apparent deflection of moving air (or water) due to Earth's rotation; deflects motion to the right in the Northern Hemisphere and to the left in the Southern Hemisphere.

geostrophic wind: Wind resulting from the balance between the pressure gradient force and the Coriolis force; blows parallel to isobars; approximates free-tropospheric wind.

thermohaline circulation: Large-scale ocean circulation driven by density differences due to temperature (thermo) and salinity (haline) variations; the global ocean conveyor belt; turnover time $\approx$1000 years.

P-wave: Primary compressional seismic wave; particle motion parallel to propagation; travels through solids and liquids; fastest seismic wave type.

S-wave: Secondary shear seismic wave; particle motion perpendicular to propagation; travels through solids only ($G = 0$ in fluids); slower than P-waves.

Mohorovičić discontinuity (Moho): The boundary between Earth's crust and mantle, identified by the increase in seismic wave speed; at $\approx$30–70 km depth under continents, $\approx$7 km under oceans.

geodynamo: The self-sustaining process by which convective motion of conducting fluid in Earth's outer core generates the planetary magnetic field through electromagnetic induction.

solar constant (S_0): The solar irradiance at Earth's mean orbital distance; $\approx$1361 W/m²; the power per unit area on a surface perpendicular to the solar beam at the top of the atmosphere.

albedo (α): The fraction of incident solar radiation reflected by a surface or planet; Earth's mean planetary albedo $\approx$0.30 (30% reflected); ice/snow $\approx$0.80; ocean $\approx$0.06.

effective temperature (T_e): The temperature a planet would need to radiate at to balance incoming solar radiation; $T_e = [S_0(1-\alpha)/(4\sigma)]^{1/4} \approx 255$ K for Earth.

climate sensitivity: The equilibrium surface warming for a doubling of atmospheric CO_2; likely range 2.5–4.0°C; best estimate $\approx$3.0°C.

Concept Review Questions

Answer in complete sentences. No calculations required unless stated.

1. Explain why the astronauts on the International Space Station (altitude 408 km) are weightless, even though gravitational acceleration at that altitude is about 88% of its surface value.
2. Atmospheric pressure at 10 km altitude is about 26 kPa — roughly 26% of sea-level pressure. Aircraft cabins are pressurised to an equivalent altitude of about 2400 m. Explain why pressurisation is necessary and what would happen to a passenger without it.
3. A rising air parcel cools at the dry adiabatic lapse rate of 9.8°C/km. The environmental lapse rate on a given day is 7.5°C/km. Is the atmosphere stable or unstable to vertical convection? Explain using density arguments.
4. Explain why S-waves do not reach seismograph stations in the S-wave shadow zone (approximately 104°–140° from the epicentre). What does this tell us about Earth's interior?
5. Explain the difference between Earth's effective radiating temperature (−18°C) and its observed mean surface temperature (+15°C). What physical process accounts for the 33°C difference?
6. Why does a tsunami travel at about 700 km/h in the deep ocean but slow to perhaps 50 km/h as it approaches a shallow coastal shelf? What happens to its amplitude during this process?
7. Explain palaeomagnetism and how the symmetric magnetic stripes on either side of mid-ocean ridges provided evidence for plate tectonics.
8. Distinguish between positive and negative climate feedbacks. Give one example of each and explain whether it amplifies or dampens an initial temperature change.

Atmospheric Pressure Problems

9. Find the atmospheric pressure at an altitude of 3000 m using the barometric formula. (P_0 = 101 325 Pa, H = 8500 m)

10. A mountain peak at 6194 m (Denali, Alaska) experiences an atmospheric pressure of approximately 47 kPa. Find the partial pressure of oxygen at this altitude. (Oxygen is 21% of the atmosphere by volume.)
11. A weather balloon rises from sea level to 30 km. Find the ratio of the pressure at 30 km to sea-level pressure using the barometric formula. Comment on whether the isothermal assumption is accurate for this altitude range.
12. The density of air at sea level is 1.225 kg/m³. Using the ideal gas law and the barometric formula, find the air density at 5500 m altitude. (M = 0.029 kg/mol, T = 253 K at 5500 m, R = 8.314 J/mol·K)
13. A tyre is inflated to 220 kPa gauge at sea level. A driver then ascends a mountain where atmospheric pressure is 78 kPa. What is the new gauge pressure in the tyre? (Assume the tyre is rigid and air temperature is unchanged.)

Weather and Fluid Motion Analysis

14. At latitude 45°N, the Coriolis parameter $f = 2\Omega \sin(45°) = 1.03 \times 10^{-4}$ rad/s. A pressure gradient of 2.0 hPa per 200 km drives geostrophic wind. If air density is 1.1 kg/m³, find the geostrophic wind speed. (Convert pressure gradient to Pa/m.)
15. Air rises in a thunderstorm updraft at 20 m/s. At what height does it reach the tropopause (12 km) assuming constant updraft speed? Find the kinetic energy per kilogram of the updraft and compare it to the gravitational potential energy gained.
16. A raindrop of radius 1.5 mm falls through air. Find its terminal velocity. (Drag coefficient $C_D = 0.47$ for a sphere; $\rho_{water} = 1000$ kg/m³; $\rho_{air} = 1.2$ kg/m³; $C_D A = \pi r^2 \times 0.47$)
17. A cold front advances at 25 km/h and encounters warm air with a temperature of 22°C and dew point of 18°C. The relative humidity is approximately 80%. Explain qualitatively (using density arguments and the lapse rate) why vigorous thunderstorms develop along the cold front boundary.

Seismic Wave Exercises

18. A P-wave travels through the crust at 6.0 km/s and through the upper mantle at 8.1 km/s. If P-waves from an earthquake arrive at a station 1200 km away 3

minutes 10 seconds after the earthquake, and S-waves (v_S = 3.5 km/s through crust) arrive later, find the P-to-S arrival time difference.

19. The density of the outer core is approximately 11 000 kg/m^3 and its bulk modulus K $\approx$ 640 GPa. Find the P-wave speed in the outer core. Then explain why S-wave speed is zero in the outer core.
20. An earthquake at an epicentral distance of 8800 km produces a P-wave arrival time of 13 minutes 20 seconds after the event. Find the average P-wave speed over this path. Why does the actual travel time differ from a simple distance/speed calculation using surface crustal velocities?
21. Three seismograph stations are located at distances of 450 km, 600 km, and 780 km from an earthquake epicentre. The P-wave speed is 6.2 km/s. Calculate the expected P-wave arrival times at each station relative to the origin time. If the absolute arrival times are known, explain how triangulation determines the epicentre.

Earth Energy Balance Problems

22. Calculate Earth's effective radiating temperature using S_0 = 1361 W/m^2, α = 0.30, and $\sigma = 5.67 \times 10^{-8}$ W/m^{2}K^4. Compare to the observed mean surface temperature of 288 K.
23. If Earth's albedo decreased from 0.30 to 0.27 (e.g., due to ice loss), find the new effective radiating temperature. By how much does T_e change? (Use the formula from Section 55.11.)
24. The Sun emits radiation as a blackbody at T = 5778 K. Find: (a) the wavelength of peak emission (Wien's law); (b) Earth's surface peak emission wavelength at T = 288 K. By what factor do these wavelengths differ? Why does this matter for greenhouse gas physics?
25. A climate model doubles atmospheric CO_2 and calculates a radiative forcing (additional energy trapped) of 3.7 W/m^2. Using a simplified linearised model $\Delta T \approx \Delta F / (4\sigma T_e^3)$, estimate the no-feedback temperature response. (T_e = 255 K) Compare to the actual climate sensitivity of $\approx$3°C that includes feedbacks.

Chapter 55 Checkpoint

Before moving on to Chapter 56, confirm you can do each of the following.

I can...	See Section
Describe Earth's layered structure and the physical state of each layer	55.1
Calculate g at altitude using $g(h) = g_0(R\_E/(R\_E+h))^2$	55.2
Apply the barometric formula $P = P_0e^{(-h/H)}$ to find pressure at altitude	55.3
Explain the lapse rate, DALR, and conditions for atmospheric instability	55.4
Describe Hadley, Ferrel, and Polar cells and the role of the Coriolis effect	55.5
Explain geostrophic wind and the Betz limit for wind turbines	55.6
Calculate terminal velocity of a falling raindrop and deep-water wave speed	55.7–55.8
Calculate tsunami speed from ocean depth using $v = \sqrt{gd}$	55.8
Distinguish P-waves and S-waves and explain what the S-wave shadow zone reveals	55.9
Describe the geodynamo and palaeomagnetic evidence for seafloor spreading	55.10
Calculate Earth's effective radiating temperature and explain the greenhouse effect	55.11
Distinguish positive and negative climate feedbacks with examples	55.12
Identify common mistakes in Earth-system physics problems	55.13

Continue to Chapter 56: Energy Systems →